

ECE 105: Introduction to Electrical Engineering

Lecture 8

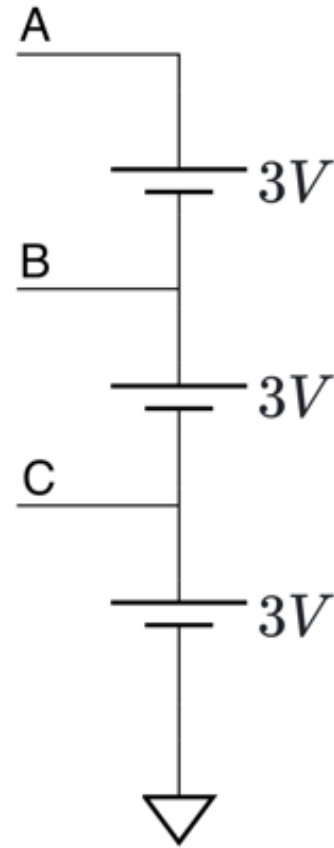
Circuit Problems - Recap

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Problem 1

Given the circuit diagram with nodes **A**, **B**, and **C**, each connected to **3V** sources as shown. Determine the voltages V_A , V_B , and V_C at the respective nodes with respect to the ground.

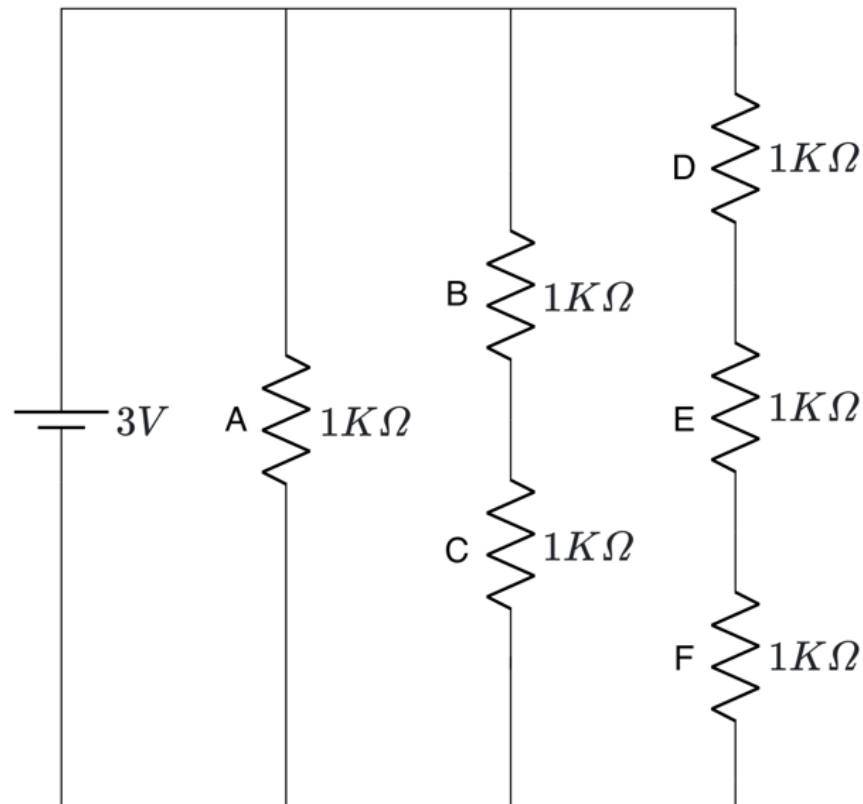


Voltage divider

Current divider

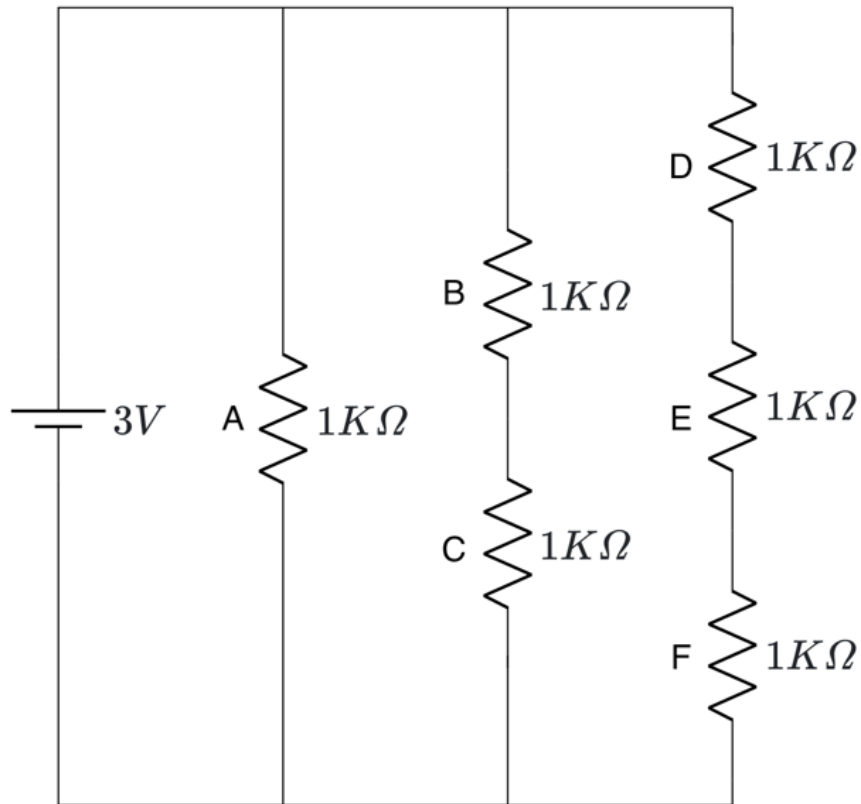
Problem 2

Given the circuit with six resistors connected to a **3V** voltage source as shown. Find voltages across the resistors (V_A , V_B , V_C , V_D , V_E , and V_F) and current through resistors (I_A , I_B , I_C , I_D , I_E , and I_F).



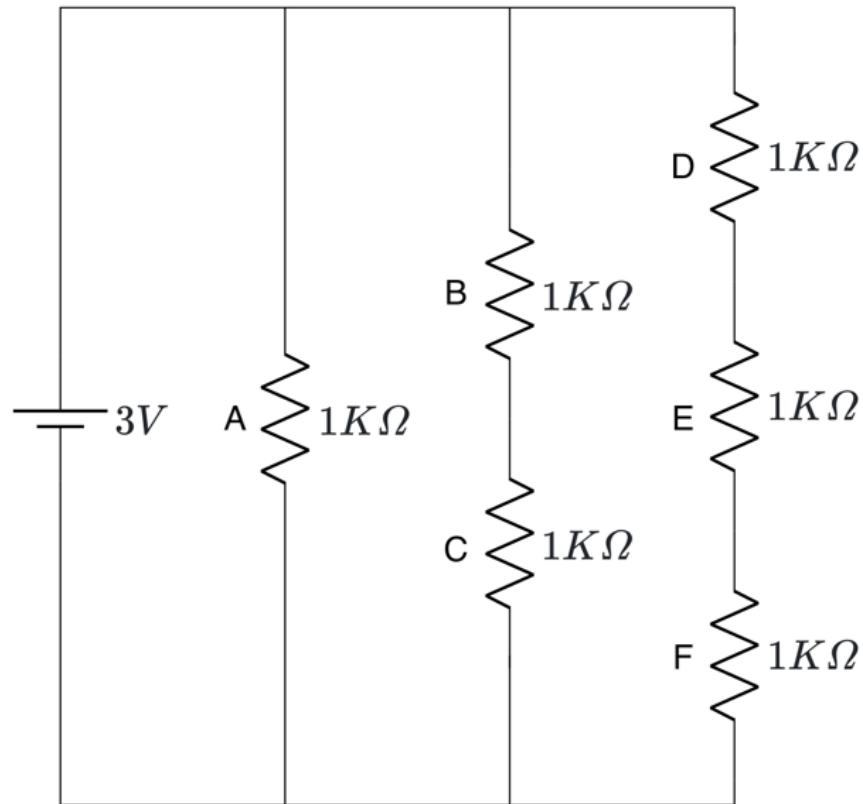
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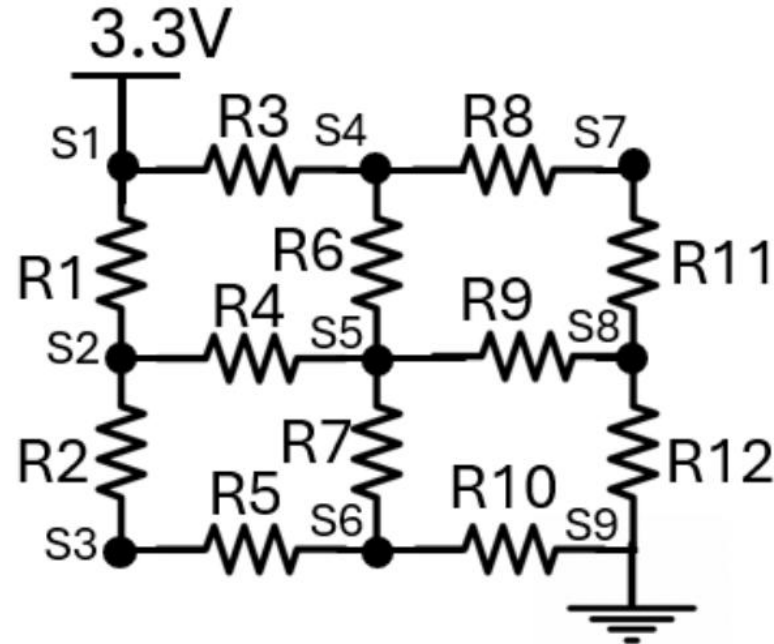
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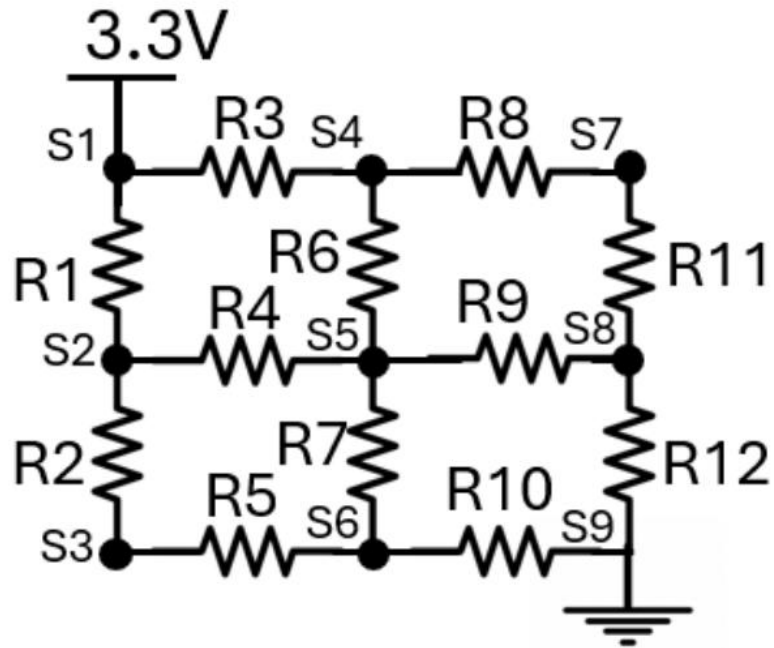
Problem 3

Given the circuit diagram with nodes S_1 to S_9 , S_1 is connected to **3.3V** and GND (0V). Determine the voltages S_1 to S_9 at the respective nodes with respect to the ground if R_1 to $R_{12} = R$



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Problem 3

Given the circuit diagram with nodes **S₁ to S₉**, **S₁** is connected to **3.3V** and GND (0V). Determine the voltages **S₁ to S₉** at the respective nodes with respect to the ground if **R₁ to R₁₂ = R**

